Research Project / Data Analysis

[Bangladesh Inflation Analysis (2010–2024)]

A Research Report on Inflation Trends in Bangladesh

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□ *Bangladesh Inflation Analysis (2010–2024)*

1. *Introduction*

Inflation is a common economic phenomenon that refers to the gradual increase in the general price level of goods and services over time. When inflation rises, the purchasing power of money decreases, which directly affects people's daily lives. For a developing country like Bangladesh, maintaining a stable inflation rate is very important for economic growth and overall economic stability.
Over the past decade, Bangladesh has experienced different phases of inflation due to both internal and external factors. This report analyzes the inflation trend in Bangladesh from 2010 to 2024 using statistical tools and graphical analysis.

2. *Objectives of the Study*

The objectives of this study are:
To examine the trend of inflation in Bangladesh
To calculate average, variance, and standard deviation
To analyze year-to-year changes in inflation
To identify factors affecting inflation
To suggest measures to control inflation

3. *Data Source and Methodology*

This study is based on secondary data collected from reliable sources such as Bangladesh Bureau of Statistics (BBS), World Bank, and IMF reports.

Data analysis was performed in Microsoft Excel.
Different statistical tools have been used for analysis, including:

- Average Inflation: The mean value of inflation over the years.

- Maximum Inflation: The highest inflation recorded in the dataset.

- Minimum Inflation: The lowest inflation recorded.

- Variance and Standard Deviation: These show how much inflation fluctuates.

- Year-over-Year Growth: Shows how inflation changes from one year to the next.

- Moving Average (3-year): Helps to understand the overall trend by smoothing short-term fluctuations.

Graphical analysis is also used to understand the trend clearly.

The final report was prepared using Microsoft word.

4. *Analysis and Interpretation*

Descriptive Statistics:

The average inflation rate during the period 2010–2024 is approximately 7.21%, which indicates a moderate level of inflation over the period.

- The maximum inflation is about 11.4%, observed in the early years.
- The minimum inflation is around 5.51%, showing the lowest price stability period.

The standard deviation is about 1.97, suggesting that inflation has not fluctuated widely. Although there are some variations, overall inflation remained within a reasonable range.

Trend of Inflation Over Time:

To better understand the overall pattern of inflation, a graphical representation is used below.

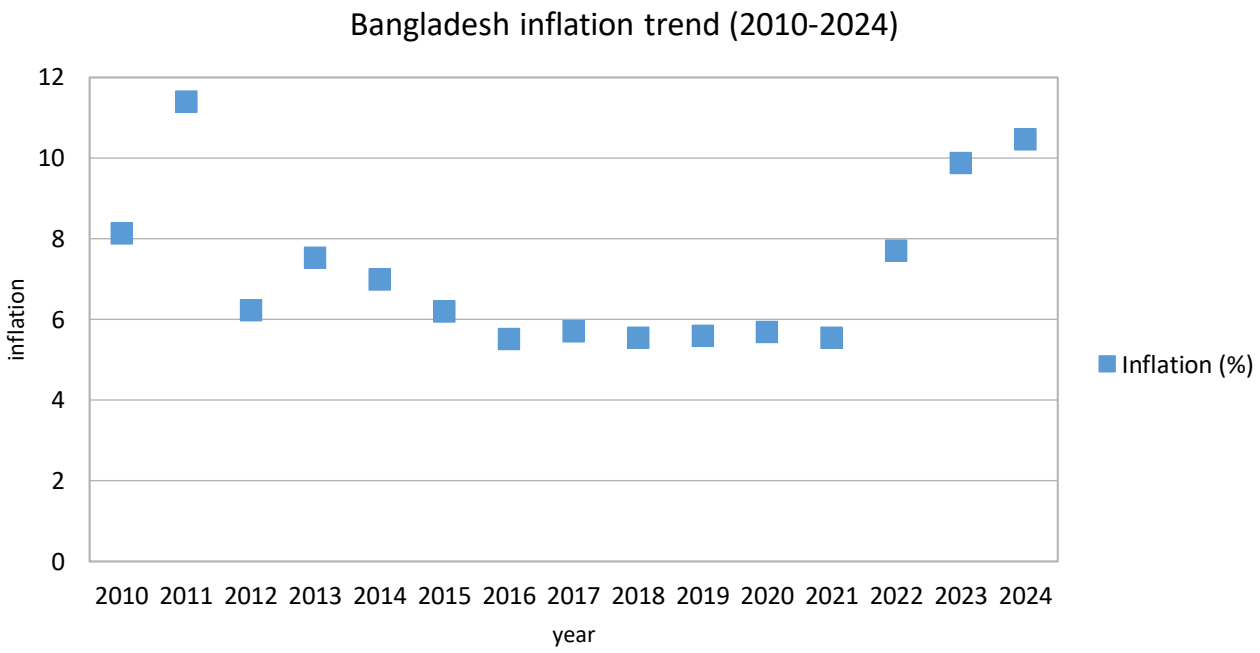

Figure 1: Inflation Trend in Bangladesh (2010–2024)

Here,
The graph shows that inflation in Bangladesh has changed over time rather than following a fixed pattern. In 2011, inflation reached a peak level of around 11.4%, which was the highest during the study period.
After that, inflation gradually declined and remained relatively stable between 2013 and 2021, mostly within the range of 5% to 6%. This period can be considered a phase of economic stability.
However, from 2022 onwards, inflation started to rise again. The increase became more noticeable in 2023 and 2024, indicating growing economic pressure.
Overall, the graph reflects a stable period followed by a recent upward trend in inflation.

Year-on-Year Growth Analysis

The year-on-year growth rates indicate that inflation did not change uniformly. In some years, inflation decreased compared to the previous year, while in other years it increased.
The average growth rate is approximately 4.29%, showing moderate but inconsistent changes over time. This suggests that inflation is influenced by both short-term and long-term factors.

Moving Average Analysis

The 3-year moving average helps to remove short-term fluctuations and provides a clearer view of the trend.
It shows that inflation remained stable during the middle years but started increasing after 2021. This indicates that the recent rise in inflation is not temporary but part of a broader trend.

5. *Factors Affecting Inflation*

Several factors have influenced inflation in Bangladesh:
Global factors such as rising fuel and food prices
Impact of the COVID-19 pandemic on supply chains
Increase in import costs
Currency depreciation
Demand and supply imbalance in the domestic market
These factors together have contributed to changes in inflation over time.

6. *Conclusion:*

The analysis shows that inflation in Bangladesh has experienced both stability and fluctuation over the study period. While inflation remained stable for several years, it has increased significantly in recent times. Although the average inflation rate is moderate, the recent upward trend is a matter of concern and requires proper attention.

7. *Recommendations:*

To control inflation, the following measures can be taken:
- The government should control the prices of essential goods.
- Strong monitoring of the market is necessary to avoid price manipulation.
- Monetary policy should be improved to control inflation.
- Supply chain management should be strengthened.
-Increasing domestic production
-Reducing dependency on imports
-Maintaining a stable exchange rate
-Providing support to low-income groups

8. *Final Comment:*

This analysis helps to understand the pattern and behavior of inflation in Bangladesh. It also highlights the importance of proper economic policies to maintain price stability.

9. *References*

Bangladesh Bureau of Statistics (BBS)
World Bank
International Monetary Fund (IMF)
Bangladesh Economic Review